

Unit 3 - Types of Random Processes

* WSS and SSS

A random process $\{X(t)\}$ is a wide sense stationary if the following conditions are satisfied.

1) $E[X(t)] = \text{a constant}$

2) $E[X(t) \cdot X(t+\tau)]$ a function of τ ($t_1 - t_2$)

Q.

A random process $X(t)$ is defined by

$X(t) = \cos \Omega t$ where Ω is a uniform random variable in $[0, \omega_0]$.

pdf = $f_{\Omega}(\omega) = \frac{1}{\omega_0}$ $0 \leq \omega \leq \omega_0$

$E[X(t)] = E(\cos \Omega t)$

$= \int_0^{\omega_0} \cos \Omega t \cdot f(\omega_0) \cdot d\Omega$

$= \frac{1}{\omega_0} \left[\sin \Omega t \right]_0^{\omega_0}$

$= \frac{1}{\omega_0} [\sin \omega_0 t - \sin(0)]$

$= \frac{\sin \omega_0 t}{\omega_0}$

$E(X(t))$ is not a constant, but is a function of t , $X(t)$ is not stationary.

Q. A random process is $X(t) = \cos(t + \phi)$ where ϕ is a R.V.

$$\text{PDF} = f(\phi) = \frac{1}{\pi}, \quad -\frac{\pi}{2} < \phi < \frac{\pi}{2}$$

$$E[X(t)] = E[\cos(t + \phi)]$$

$$= \int_{-\pi/2}^{\pi/2} \cos(t + \phi) \cdot \frac{1}{\pi} \cdot d\phi$$

$$= \frac{1}{\pi} \left[\sin(t + \phi) \right]_{-\pi/2}^{\pi/2}$$

$$= \frac{1}{\pi} \left[\sin(t + \pi/2) - \sin(t - \pi/2) \right]$$

$$= \frac{1}{\pi} [\cos(t) - (-\cos(t))]$$

$$= \frac{2 \cos t}{\pi}$$

$\therefore E(X(t))$ is not a constant, but a function of t
 Hence the process is not stationary.

Q. A Random process is given by $X(t) = a \sin(\omega_0 t)$ where ω_0 is a constant and a is a R.V. uniformly distributed over $[-1, 1]$. Prove that the process is not stationary.

$$\text{PDF} = f(a) = \frac{1}{2}, \quad -1 < a < 1$$

$$\begin{aligned}
 1) \quad E[X(t)] &= E[a \sin(\omega_0 t)] \\
 &= \int_{-1}^1 a \sin(\omega_0 t) \cdot \frac{1}{2} da
 \end{aligned}$$

$$= \frac{\sin \omega_0 t}{2} \int_{-1}^1 a \, da$$

$$= \frac{\sin \omega_0 t}{2} \left[\frac{a^2}{2} \right]_{-1}^1$$

$$= \frac{\sin \omega_0 t}{4} \cdot [(1) - (1)]$$

$$= 0 \quad \rightarrow \text{a constant.}$$

$$2) R_x(\tau) = E[X(t) \cdot X(t+\tau)]$$

$$= E[a \sin \omega_0 t \cdot a \sin \omega_0 (t+\tau)]$$

$$= E[\sin \omega_0 t \cdot \sin \omega_0 (t+\tau)] \cdot E[a^2]$$

$$= \sin \omega_0 t \cdot \sin \omega_0 (t+\tau) \cdot \frac{1}{2} \int_{-1}^1 a^2 \, da$$

$$= \sin \omega_0 t \cdot \sin \omega_0 (t+\tau) \cdot \frac{1}{2} \cdot \left[\frac{a^3}{3} \right]_{-1}^1$$

$$= \sin \omega_0 t \cdot \sin \omega_0 (t+\tau) \cdot \left(\frac{1}{3} - \left(-\frac{1}{3} \right) \right)$$

$$= \sin \omega_0 t \cdot \sin \omega_0 (t+\tau) \cdot \frac{2}{3}$$

$$= \frac{\sin \omega_0 t \cdot \sin \omega_0 (t+\tau)}{3}$$

$\therefore R_x(\tau)$ is not purely a function of τ ,
 $\therefore X(t)$ is not stationary.

Q. Show that the random process given by
 $X(t) = A \cos(\omega_0 t + \theta)$
 where A and ω_0 are constants and
 θ is uniformly distributed over $[0, 2\pi]$ is
 a wide sense stationary.

$$pdf = f(\theta) = \frac{1}{2\pi}, \quad 0 < \theta < 2\pi$$

$$\begin{aligned}
 1) \quad E[X(t)] &= E[A \cos(\omega_0 t + \theta)] \\
 &= \int_0^{2\pi} A \cos(\omega_0 t + \theta) f(\theta) d\theta \\
 &= \int_0^{2\pi} A \cos(\omega_0 t + \theta) \cdot \frac{1}{2\pi} d\theta \\
 &= \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{2\pi} \\
 &= \frac{A}{2\pi} \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t + 0) \right] \\
 &= \frac{A}{2\pi} \left[\sin \omega_0 t \cos 2\pi + \cos \omega_0 t \sin 2\pi - \sin \omega_0 t \cos 0 + \cos \omega_0 t \sin 0 \right] \\
 &= \frac{A}{2\pi} \left[2 \cos \omega_0 t \sin 2\pi \right] \rightarrow 0.
 \end{aligned}$$

$\sin 2\pi = 0$, $\cos 0 = 1$, $\sin 0 = 0$, $\cos 2\pi = 1$.
 Hence, $E[X(t)] = 0$, a constant.

$$\begin{aligned}
 2) \quad R_X(t) &= E[A \cos(\omega_0 t + \theta) \cdot A \cos(\omega_0 (t + \tau) + \theta)] \\
 &= A^2 E[\cos(\omega_0 t + \theta) \cos(\omega_0 (t + \tau) + \theta)] \\
 &= A^2 E[\cos(\omega_0 t + \theta + \omega_0 (t + \tau) + \theta) + \cos(\omega_0 t + \theta - \omega_0 (t + \tau) - \theta)] \\
 &= \frac{A^2}{2} E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta) + \cos(\omega_0 \tau)] \\
 &= \frac{A^2}{2} E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)] + \frac{A^2}{2} E[\cos(\omega_0 \tau)]
 \end{aligned}$$

(1) Ensemble average
 (2) Time average

$$E[\cos(2\omega_0 t + \omega_0 \tau + 2\theta)] = \int_0^{2\pi} \cos(2\omega_0 t + \omega_0 \tau + 2\theta) \cdot \frac{1}{2\pi} d\theta = 0$$

$$E[\cos(\omega_0 \tau)] = \int_0^{2\pi} \cos(\omega_0 \tau) \cdot \frac{1}{2\pi} d\theta = \cos(\omega_0 \tau)$$

* Evolutionary Process

A process which is not stationary in any sense is called evolutionary process. In other words we can say that a random process is evolutionary if the mean and the variance are a function of t i.e. they are dependent on t .

* Ergodicity

Time average: $\bar{X}_t = \frac{1}{2T} \int_{-T}^T X(t) dt$

A random process is ergodic when time average is equal to ensemble average.

$$\frac{1}{2T} \int_{-T}^T X(t) dt \rightarrow \mu \text{ as } T \rightarrow \infty \text{ where } \mu = \text{ensemble average} = E[X(t)]$$

Q. A random process is given by $X(t) = A \cos(\omega_0 t + \theta)$ where θ is a R.V. uniformly distributed over $(0, 2\pi)$. Show that the process is mean-ergodic.

① Ensemble average

$$\begin{aligned} E[X(t)] &= E[A \cos(\omega_0 t + \theta)] \\ &= \int_0^{2\pi} A \cos(\omega_0 t + \theta) \cdot f(\theta) d\theta \\ &= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \theta) d\theta \\ &= \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{2\pi} \end{aligned}$$

$$\phi = A \left[\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t) \right]$$

$$\omega = \frac{A}{2\pi} \left[\sin(\omega_0 t) - \sin(\omega_0 t) \right]$$

$$= \frac{A}{2\pi} (0) = \underline{0}$$

$$\phi = A \cos(\omega_0 T + \theta)$$

$$\sin(-x) = -\sin(x)$$

$$\bar{X}_T = \frac{1}{2T} \int_{-T}^T x(t) dt$$

$$= \frac{1}{2T} \int_{-T}^T A \cos(\omega_0 t + \theta) dt$$

$$= \frac{A}{2T} \int_{-T}^T \cos(\omega_0 t + \theta) dt$$

$$= \frac{A}{2T} \left[\frac{\sin(\omega_0 t + \theta)}{\omega_0} \right]_{-T}^T$$

$$= \frac{A}{2T\omega_0} \left[\sin(\omega_0 T + \theta) - \sin(-\omega_0 T + \theta) \right]$$

$$= \frac{A}{2T\omega_0} \left[\sin(\omega_0 T + \theta) + \sin(\omega_0 T - \theta) \right]$$

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T x(t) dt = 0$$

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Q. A random process given by $X(t) = \cos(t + \phi)$ where ϕ is a R.V. uniformly dist. over $(0, 2\pi)$

- (i) Show that the process is stationary in wide sense.
 (ii) Show that the process is mean ergodic.

(i) pdf = $f(\phi) = \frac{1}{2\pi - 0} = \frac{1}{2\pi} \dots 0 < \phi < 2\pi$

$$\begin{aligned}
 E[X(t)] &= E[\cos(t + \phi)] \\
 &= \int_0^{2\pi} \cos(t + \phi) \cdot f(\phi) \cdot d\phi \\
 &= \frac{1}{2\pi} \left[\sin(t + \phi) \right]_0^{2\pi} \\
 &= \frac{1}{2\pi} [\sin(t + 2\pi) - \sin(t)] \\
 &= \frac{1}{2\pi} [\sin(t) - \sin(t)] = \frac{0}{2\pi} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 E[X(t) \cdot X(t + \tau)] &= E[\cos(t + \phi) \cdot \cos(t + \tau + \phi)] \\
 &= \frac{1}{2} E[\cos(2t + \tau + 2\phi) + \cos(\tau)] \\
 &= \frac{1}{2} \cos(\tau) + \frac{1}{2} \int_0^{2\pi} \cos(2t + \tau + 2\phi) \cdot \frac{1}{2\pi} d\phi \\
 &= \frac{1}{2} \cos(\tau) + \frac{1}{4\pi} \int_0^{2\pi} \cos(2t + \tau + 2\phi) d\phi \\
 &= \frac{1}{2} \cos(\tau) + \frac{1}{4\pi} \left[\sin(2t + \tau + 2\phi) \right]_0^{2\pi} \\
 &= \frac{1}{2} \cos(\tau) + \frac{1}{8\pi} [\sin(2t + \tau + 2\pi) - \sin(2t + \tau)]
 \end{aligned}$$

$$= \frac{\cos(\tau)}{2} + \frac{1}{8\pi} [\sin(2t + \tau) - \sin(2t + \tau)]$$

$$= \frac{\cos(\tau)}{2} + \frac{1}{8\pi} X(0)$$

$$= \frac{\cos(\tau)}{2}$$

$$\text{Time avg} = \bar{X}_T = \frac{1}{2T} \int_{-T}^T X(t) dt$$

$$= \frac{1}{2T} \int_{-T}^T \cos(t + \phi) dt$$

$$= \frac{1}{2T} [\sin(t + \phi)]_{-T}^T$$

$$= \frac{1}{2T} [\sin(T + \phi) - \sin(-T + \phi)]$$

$$= \frac{1}{2T} [\sin(T + \phi) + \sin(T - \phi)]$$

Since $\sin(T \pm \theta)$ is bounded as $T \rightarrow \infty$,

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T X(t) dt = 0$$

Since Time avg = Ensemble avg = 0,
 $X(t)$ is mean ergodic.

Q. A random process is given by ~~$x(t)$~~

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

where A and B are R.V. such that

$$E[A] = 0$$

$$E[B] = 0$$

$$E[AB] = 0$$

$$E[A^2] = E[B^2]$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

Since $x(t)$ is bounded $\therefore$ $x(t)$ is bounded

$$0 \leq x(t) \leq 1$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

$$x(t) = A \cos(\omega t) + B \sin(\omega t)$$

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* Markov Chain

A random process in which the set of states, space S and the set of parameter T are discrete taking only a finite or countable number of finite values in which the probability p_{ij} , that the process will go from i th state to j th state in the next step depends only on the present state and not on how it arrived at that state is called Markov Chain of the first order.

$$P [X_{n+1} = j \mid X_0 = i_0, X_1 = i_1, X_2 = i_2, \dots, X_n = i_n]$$

$$= P [X_{n+1} = j \mid X_n = i_n]$$

$X_0, X_1, \dots, X_{n-1} \rightarrow$ past values

$X_n \rightarrow$ present value

$X_{n+1} \rightarrow$ future value

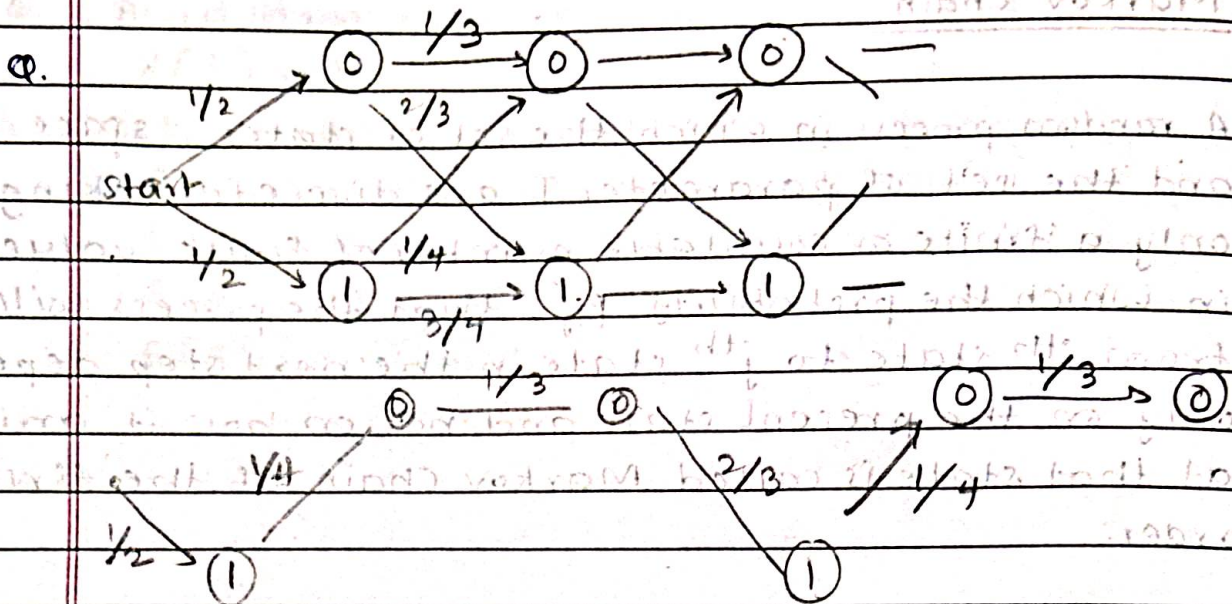
One Step Transition Probability Matrix

		States of X_{n+1}			
		0	1	2	...
States of X_n	0	p_{00}	p_{01}	p_{02}	$\dots$
	1	p_{10}	p_{11}	p_{12}	$\dots$
	2	p_{20}	p_{21}	p_{22}	$\dots$
	:	$\dots$	$\dots$	$\dots$	$\dots$

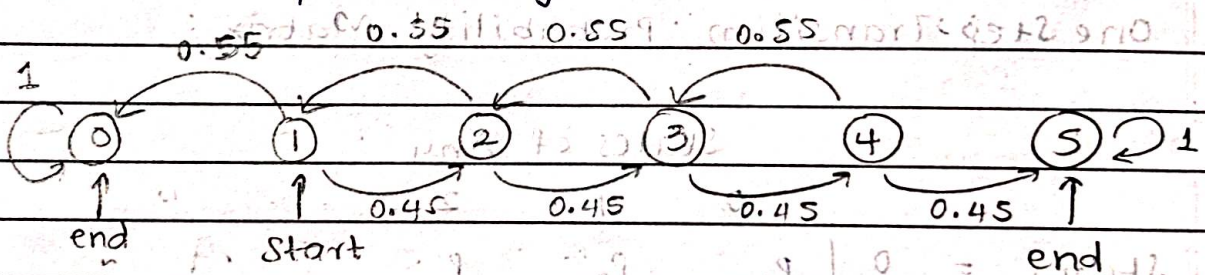
n-step TPM

Rules:

1. $p_{ij} \geq 0, \quad i, j \geq 0$
2. $\sum_{j=0}^{\infty} p_{ij} = 1, \quad i = 0, 1, 2, \dots$



Q. Suppose in a gambling game, a gambler wins Rs. 1 with probability 0.45 and loses Rs. 1 with probability 0.55 at any state. He starts the game with Rs. 1 and quits it when either he has earned Rs. 4 or he is bankrupt. Write the transition probability matrix.



		X_{n+1}					
		0	1	2	3	4	5
X_n	0	0	0	0	0	0	0
	1	0.55	0	0.45	0	0	0
	2	0	0.55	0	0.45	0	0
	3	0	0	0.55	0	0.45	0
	4	0	0	0	0.55	0	0.45
	5	0	0	0	0	0	1

Q. Consider an experiment in which a rat is wandering inside a maze which has 8 compartments.

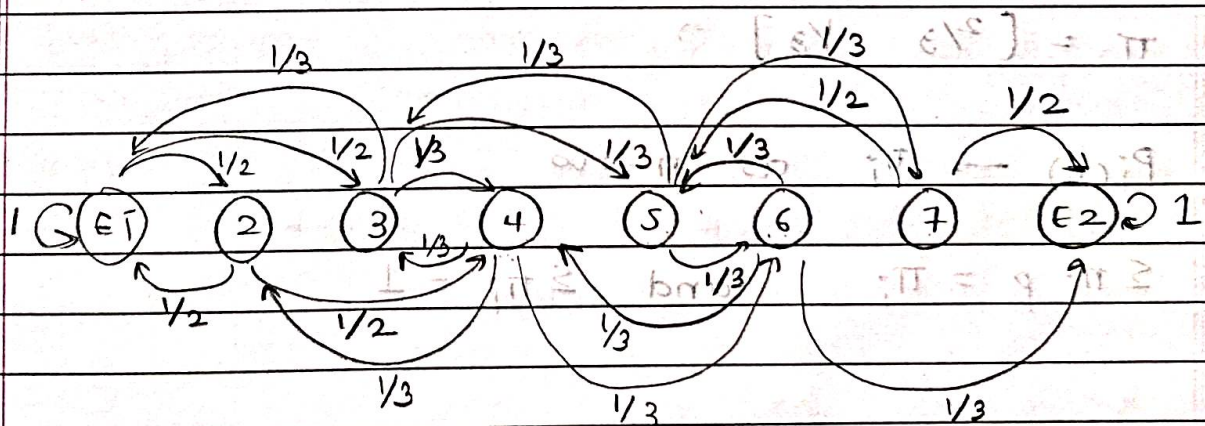
E1	3	5	7
2	4	6	E2

If the rat reaches the comp. E1, E2, it escapes.

Otherwise it moves from one compartment to the other through the rows at random. Write the TPM and transition diagram.

X_{n+1}

		E1	2	3	4	5	6	7	E2
X_n	E1	1	0	0	0	0	0	0	0
	2	1/2	0	0	1/2	0	0	0	0
	3	1/3	0	0	1/3	1/3	0	0	0
	4	0	1/3	1/3	0	1/3	0	0	0
	5	0	0	1/3	0	0	1/3	1/3	0
	6	0	0	0	1/3	1/3	0	0	1/3
	7	0	0	0	0	1/2	1/2	0	1/2
	E2	0	0	0	0	0	0	0	1



* Initial Conditions

$$Q. \begin{bmatrix} 0.4 & 0.4 & 0.2 \\ 0.1 & 0.4 & 0.5 \\ 0.0 & 0.6 & 0.4 \end{bmatrix} M_1 \quad \begin{matrix} \mu(1) = 0.2 \\ \mu(2) = 0.5 \\ \mu(3) = 0.3 \end{matrix} M_2$$

Find the probabilities of process at time t_1

$$M_2 \cdot M_1 = \begin{bmatrix} 0.13 & 0.46 & 0.41 \end{bmatrix}$$

* Steady State Probabilities (Limiting probabilities)

$$p = \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix}$$

$$p^2 = \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{bmatrix}$$

$$p^3 = \begin{bmatrix} 0.67 & 0.33 \\ 0.66 & 0.34 \end{bmatrix} \begin{bmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{bmatrix} = \begin{bmatrix} 0.667 & 0.333 \\ 0.666 & 0.334 \end{bmatrix}$$

$$\pi = \begin{bmatrix} 2/3 & 1/3 \end{bmatrix}$$

$$p_i(n) \rightarrow \pi_i \text{ as } n \rightarrow \infty$$

$$\sum \pi_i p = \pi_i \quad \text{and} \quad \sum \pi_i = 1$$

Q. If the TPM of Markov chain is $P = \begin{bmatrix} 0 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$

Find its steady state matrix. (upto power 8, $n = 8$)

$$P^2 = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 3/4 \end{bmatrix}$$

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Poisson Process

$$P[X(t) = x] = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots$$

Mean and Variance of Poisson Process

$$M = \lambda t, \quad V = \lambda t$$

Autocorrelation

$$R_X(t_1, t_2) = \lambda^2 t_1 t_2 + \lambda t_1$$

Autocovariance

$$C_X(t_1, t_2) = \lambda t_1$$

Q. If a bank receives an average 6 bad cheques per day. What are the probabilities that it will receive

- (i) 4 bad cheques on any given day
- (ii) 10 bad cheques on any two consecutive days.

(i) $\lambda = 6$ $t = 1$

$$\lambda t = 6$$

 $x = 4$

$$P[X(1) = 4] = \frac{e^{-6} (6)^4}{4!} = 0.134$$

(ii) $t = 2$

$$\lambda t = 12$$

 $x = 10$

$$P[X(2) = 10] = \frac{e^{-12} (12)^{10}}{10!} = 0.105$$

Q. Customers arrive at a counter in accordance with poisson process with mean rate of 2 per minute. Find the probability that the interval between 2 consecutive arrivals is

- (i) More than 1 minute
- (ii) Between 1 and 2 minutes
- (iii) 4 minutes or less

By the property of Poisson Process, the interval of time, t , between two consecutive customers follows exponential distribution with

$$\lambda = 2, \quad t = 1$$

$$f(t) = \lambda e^{-\lambda t} = 2e^{-2t}$$

$$(i) \quad P(t > 1) = \int_1^{\infty} 2e^{-2t} dt$$

$$= \cancel{2} \cdot \left[\frac{e^{-2t}}{-2} \right]_1^{\infty}$$

$$= - [e^{-\infty} - e^{-2}] = 0.135$$

$$(ii) \quad P(1 < t < 2) = \int_1^2 2e^{-2t} dt$$

$$= - [e^{-2t}]_1^2 = - [e^{-4} - e^{-2}]$$

$$= 0.117$$

$$(iii) \quad P(t < 4) = \int_0^4 2e^{-2t} dt$$

$$= - [e^{-2t}]_0^4 = - [e^{-8} - e^0]$$

$$= 0.999$$

Q. A radioactive substance emits particles at the rate of 6 particles per minute acc poisson process. The prob. that the emitted particles will be recorded as 0.5. Find the prob. that 10 particles will be recorded in 4 mins time period.

→

$$P[N(t)=r] = \frac{e^{-\lambda t} \cdot (\lambda t)^r}{r!}$$

$$P = \frac{e^{-(6)(0.5)(4)} \cdot (6 \times 0.5 \times 4)^{10}}{10!}$$

$$= 0.104$$